

Question number	Answer	Mark
5(a)	The only correct answer is C. <i>A is incorrect because dendrochronology is dating trees using tree rings</i> <i>B is incorrect because epigenetics studies changes in gene expression</i> <i>D is incorrect because species diversity is a measure of the number of different species in an area</i>	(1)

Question number	Answer	Additional guidance	Mark
5(b)(i)	• 0.12 (1) • mm hr⁻¹ (1)	NB If different units have been used, award a correct numerical value e.g. 0.012 cm hr⁻¹ = 2 marks	(2)

Question number	Answer	Additional guidance	Mark
5(b)(ii)	An answer that includes three of the following points: • suitable as a mean is calculated (1) • but there is no indication of sample size so may not be that suitable (1) • data looks more suitable in the first 45 hours as the error bars are small (1) • may not be suitable as the incubation temperature may be different to body temperature (1) • will not be suitable if person died more than 120 hours ago (1)	ACCEPT {reliable / accurate} as an alternative for suitable throughout ACCEPT overall the error bars are (quite) small parts of the graph where the error bars do not overlap are suitable / converse ACCEPT body temperature does not remain constant	(3)

Question number	Answer	Additional guidance	Mark
5(b)(iii)	A description that includes three of the following points: • (start the investigation with) several blowfly eggs (1) • measure the length of larvae (at intervals) over a period of 120 hours (1) • credit a named control variable appropriate to this data collection (1) • calculate mean and {standard deviation / range bars / error bars} (1)	e.g. same food supply, species of blowfly IGNORE inappropriate named control variables eg pH	(3)

Question number	Answer	Additional guidance	Mark
5(c)	An answer that includes four of the following points: THREE FROM: - body temperature is readily available information (1) {calibration curves / formulae} available to work backwards to estimate time of death (1) - but the change in body temperature is due to several factors (1) - credit example of one of these factors (1) AND - therefore of limited use unless used in conjunction with other methods (1)	ACCEPT calculations can be made to estimate time of death e.g. ambient temperature ACCEPT on its own	(4)